

MIAMA active travel and health impact methodology

Draft methods description for methodological review and early testing

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2026-09-07

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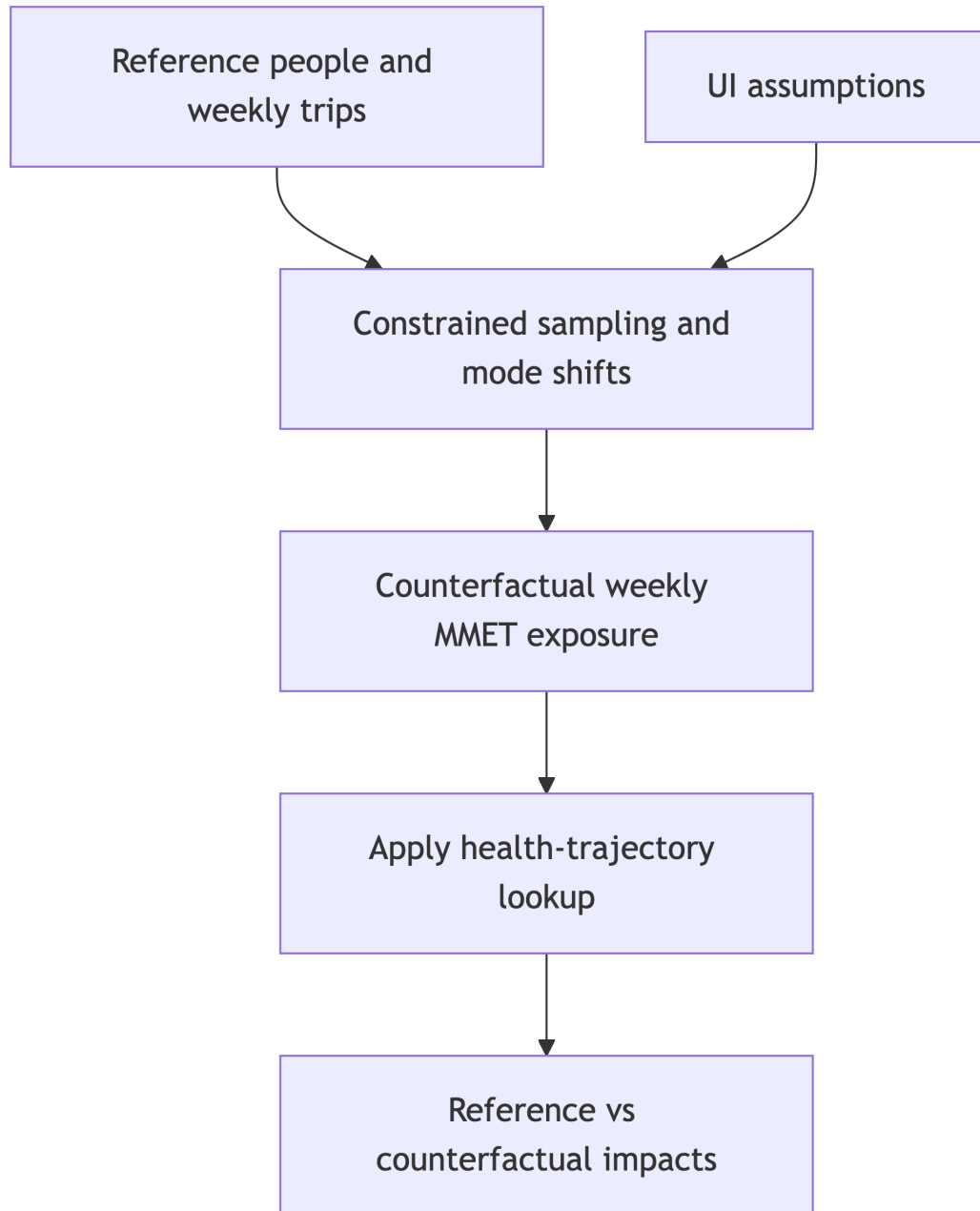
Document status

This is a methods-facing description of the behavior implemented in MIAMA-HUB, not a final model specification. **Implemented** means that the behavior is present in the current code. **Approximate** means that it is implemented through probabilistic sampling and may not exactly reproduce a requested distribution. **Planned** identifies a UI option or methodological feature that is represented in the design but does not yet change modeled outcomes.

1 Purpose and scope

MIAMA estimates how a proposed intervention may change walking, cycling, e-biking, and public-transport use and how the resulting change in physical activity may alter population health over time. The current workflow is a comparative scenario model:

1. select a reference synthetic population and its observed weekly travel;
2. translate user-entered active-travel assumptions into a counterfactual copy of those people and trips;
3. convert changed active time, including walking access to public transport, into marginal metabolic-equivalent exposure (MMETs);
4. apply the MMET change to health-model trajectories; and
5. report the difference between the reference (without-scheme) and counterfactual (with-scheme) scenarios.



The model is not a transport-demand model and does not create a new population. It reallocates active travel among the existing synthetic people and trips, with a limited mechanism for induced recreational trips. The validity of the results therefore depends on both the external assumptions supplied through the app and the representativeness of the donor people and trips available in the selected geography.

2 Reference synthetic population

2.1 Packaged geographic samples

Two independent LAD profiles are available: Leeds (5,000 people) and Manchester (10,000 people, Manchester LAD rather than Greater Manchester). Manchester is a simple random sample without replacement from 27,864 source individuals, using seed 20260908. The same selected person IDs identify the associated trip records, overall health outcomes and death-share cycle histories. The build rejects missing health histories and duplicate person/person-cycle health records. It does not rerun the health model or invent new observations.

Profiles are kept in separate directories and share common health lookup and disability-parameter files. Geographic source expansion (55.728 for Manchester) is recorded as provenance only; each selected appraisal row still represents one person. A larger profile increases the distinct donor evidence available, not the weight of each appraisal record. Profile selection occurs before loading reference data and is distinct from subsequent REF/CF appraisal sampling.

2.2 Unit of representation

The synthetic population represents approximately 5% of the source population. Each person row is currently assigned a population factor of 20 when aggregate health results are presented. Individual rows remain fixed between scenarios: investment in active travel changes behavior, not the number of residents.

The data contain two related levels:

- **Individual level:** one row per synthetic person, including geography, demographics, household characteristics, and weekly activity.
- **Trip level:** reported trips linked to people by `census_id`, including travel mode, purpose, distance, duration, and a trip weight.

Person and trip records are required for the most complete workflow. Trip rows contain person identifiers and selected attached attributes, but they do not replace the individual table: people without usable trip rows and person-level physical-activity measures still matter.

2.3 Key variables from active travel to physical activity

Domain	Principal fields	Role in the model
Identity and geography	<code>census_id</code> , <code>nts_id</code> , LAD and region identifiers	Select the appraisal population and link people, trips, and health trajectories.
Demographics	<code>age1year</code> , <code>female</code> , <code>ethnicity</code> , deprivation, limiting condition	Describe the reference population, constrain counterfactual sampling, and identify health-model strata.

Domain	Principal fields	Role in the model
Weekly activity	<code>walktime_wkhr</code> , <code>cycletime_wkhr</code> , derived e-bike and PT-access fields, <code>sport_wkhr</code>	Person-level weekly hours used to identify mode users and calculate MMET exposure.
Trip mode	<code>trip_mainmode</code> and mode-component fields	Identify mode trips and record mode switches.
Trip amount	raw and mode-specific distance and duration	Preserve plausible observed journeys and calculate added or removed active minutes.
Trip purpose	<code>trip_purpose</code> , derived <code>trip_utilitarian</code>	Distinguish travel needs from discretionary recreational travel.
Trip representation	one trip row per record; <code>weight_tripXhh</code> retained as source metadata	Define current appraisal trip counts and preserve a future route to weighted reporting.

Mode-user indicators are derived from positive mode-specific weekly time, with trip evidence used where appropriate. Active-trip indicators use the mode-specific trip component fields and main-mode coding. HUB adds explicit `user_walk`, `user_bike`, `user_ebike`, `user_pt`, `trip_activemode`, and `trip_utilitarian` fields to make these model currencies inspectable.

2.4 Weekly physical-activity exposure

The model uses the week as its base behavioral timeframe. Current conversion factors are configurable and equal to:

Activity	MMET per hour
Walking	2.5
Cycling	5.8
E-biking	5.8 (cycling proxy)
PT access walking	2.5
Other vigorous activity	7.0

For person-level changes, weekly hours are converted to MMET-week-hours and added to or subtracted from the person's reference MMET value. For an independently specified trip target, active minutes are compared between reference and counterfactual trips, divided by 60, and multiplied by the mode-specific intensity. User-triggered trip changes are excluded from this second calculation because their exposure is already represented by the person-level activity change.

This separation prevents the same behavioral change from being counted once as weekly user activity and again as changed trip minutes.

3 From physical activity to health trajectories

MIAMA-HM provides reference person-cycle health outcomes and a piecewise lookup describing the change in each outcome per additional MMET. HUB does not rerun the underlying microsimulation. It applies the scenario logic to the processed health-model outputs.

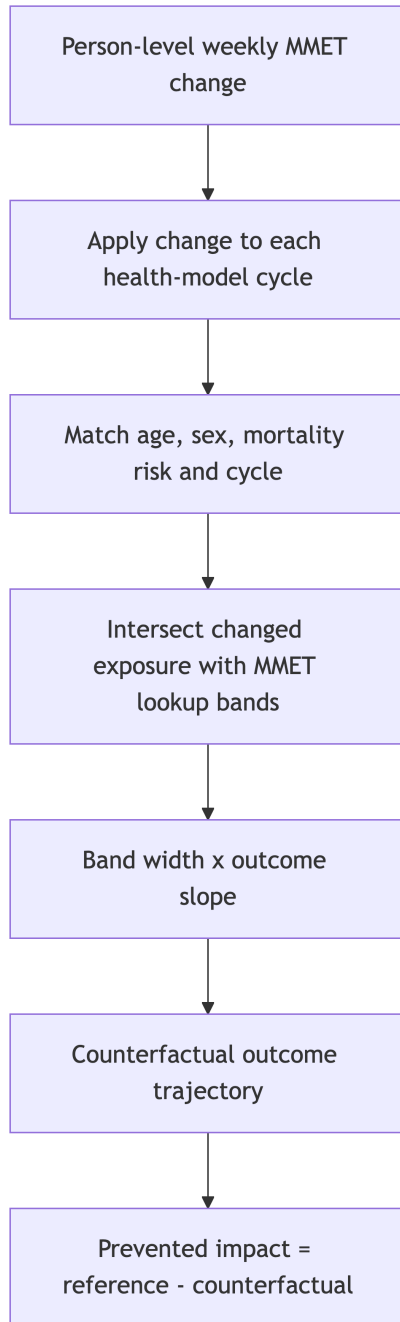
For each synthetic person:

$$\Delta MMET_i = MMET_{i,cf} - MMET_{i,ref}$$

Under the currently implemented **long-term** assumption, the same individual MMET change is applied in every modeled cycle:

$$MMET_{i,t,cf} = MMET_{i,t,ref} + \Delta MMET_i$$

The changed MMET interval is intersected with the applicable lookup bands for the person's age, sex, mortality-risk decile, and model cycle. For each overlap, the model multiplies the interval width by the band-specific outcome slope and sums across bands. Counterfactual outcomes equal reference outcomes plus this technical delta.



The internal `d_*` fields use `counterfactual - reference`. Presentation fields reverse the sign so that a positive prevented value denotes a health gain. The configured assessment period is currently 40 years, excluding cycle 0 from impact totals. Population totals apply the synthetic-person factor described above.

3.1 Health-adjusted life years

Healthy life years (HLYs) and health-adjusted life years (HALYs) are separate measures. HALYs are calculated after the MMET lookup has produced the counterfactual incidence trajectories. For each

person and year, HUB reconstructs prevalence from incidence, deaths, disease-specific death share, and, for depression, remission/exits. It does this independently for reference and counterfactual trajectories.

Life years are first adjusted for the age/sex residual prevalence of years lived with disability outside the explicitly modelled diseases. They are then adjusted for reconstructed disease prevalence multiplied by age-, sex-, and disease-specific disability weights that account for comorbidity:

$$HALY_{i,t} = LY_{i,t}(1 - pYLD_{i,t}) \left(1 - \sum_d prevalence_{i,t,d} dw_{i,t,d}^{adj} \right)$$

If nobody remains alive in a person-cycle, both prevalence-related disability and HALYs are zero. Cycles beyond the maximum age covered by the parameter tables are outside the HALY model and are omitted from HALY summaries, while their underlying health-outcome records remain available.

The technical HALY delta is counterfactual minus reference, so a positive value already represents a health gain. This differs from adverse incidence outcomes, where positive presentation benefits require reversing the technical delta.

The health outcome set includes mortality, life-year, HALY, and disease-incidence measures supplied by the death-share health-model artifacts. Effects attributed to walking or cycling are currently obtained after the combined health run by allocating impacts in proportion to each person’s signed mode-specific MMET contribution. This is an attribution of a combined nonlinear calculation, not an independent mode-specific rerun of the health model.

4 Ways to specify counterfactual active travel

4.1 Reference defaults

After geography and assessed modes are selected, HUB loads and filters the synthetic population and fills the UI profile’s reference `default_value` fields. It calculates reference values for all supported modes and common units up front; later UI changes can therefore switch displays without reloading the large data.

Counterfactual controls with a matching reference control receive the reference value as their starting `default_value`. This represents a no-change scenario; it does not mark the control as user-filled and does not overwrite a submitted counterfactual value.

For the advanced population total and per-mode counts, the initial geography-derived value is also stored as `additional_data$default_value_backup` on both reference and counterfactual fields. Refinement controls may change the live `default_value`; the backup retains the unrefined starting point for restoration. Selecting a new geography and rebuilding reference defaults refreshes the backup.

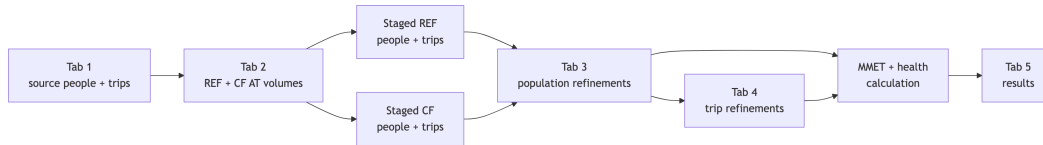
Reference values include user and trip counts, mode share, average distance and duration, and compact five-category distributions. Weekly values are canonical. Trip counts can be converted between day, week, and year; user-count conversion currently uses a factor of one because repeat-user behavior outside the observed week is not identified by the data.

Two Tab 3 selectors also receive derived `additional_data`. Separate `ref` and `cf` payloads hold absolute synthetic-row counts for each staged scenario. For each configured age group and PA group, HUB records the number of people in the category (`pop_tot`) and the numbers identified as walking, cycling, e-biking, or walk-to-public-transport users. Age groups use `age1year` and the configured age cut points. PA groups use weekly MMET-hours from `mmets` or `mmet_wkhr`; if neither is present, HUB reconstructs the value from walking, cycling, and sport hours and their configured intensities. Mode-user status is identified from person-level mode duration where available, with trip evidence as fallback. Missing evidence is retained as `NA`, not interpreted as no users. The categories exhaust the staged population: age includes under-18 and other/unknown groups, while PA includes an unknown group. These metadata populate labels/context for category choices while leaving the selector's categorical `default_value` unchanged. They are counts in the filtered synthetic profile; population expansion factors are applied to health results rather than to these selector metadata.

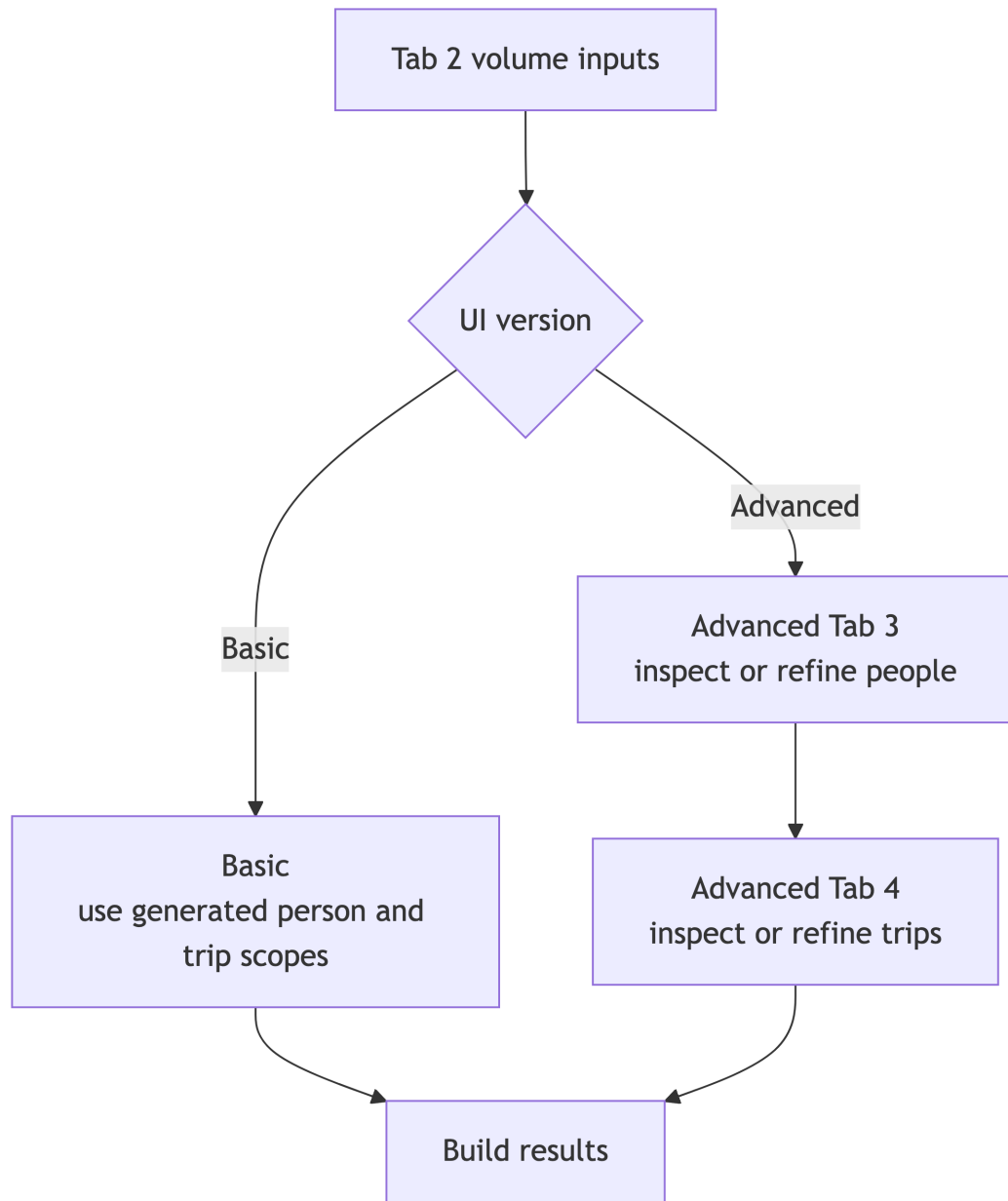
4.2 Continuity from Tab 2 to Tabs 3 and 4

4.2.1 Workflow overview

The diagrams below separate four ideas that can otherwise be conflated: the geographic donor data, the REF/CF volumes requested in Tab 2, the people represented by those volumes in Tab 3, and the trips finally selected in Tab 4.

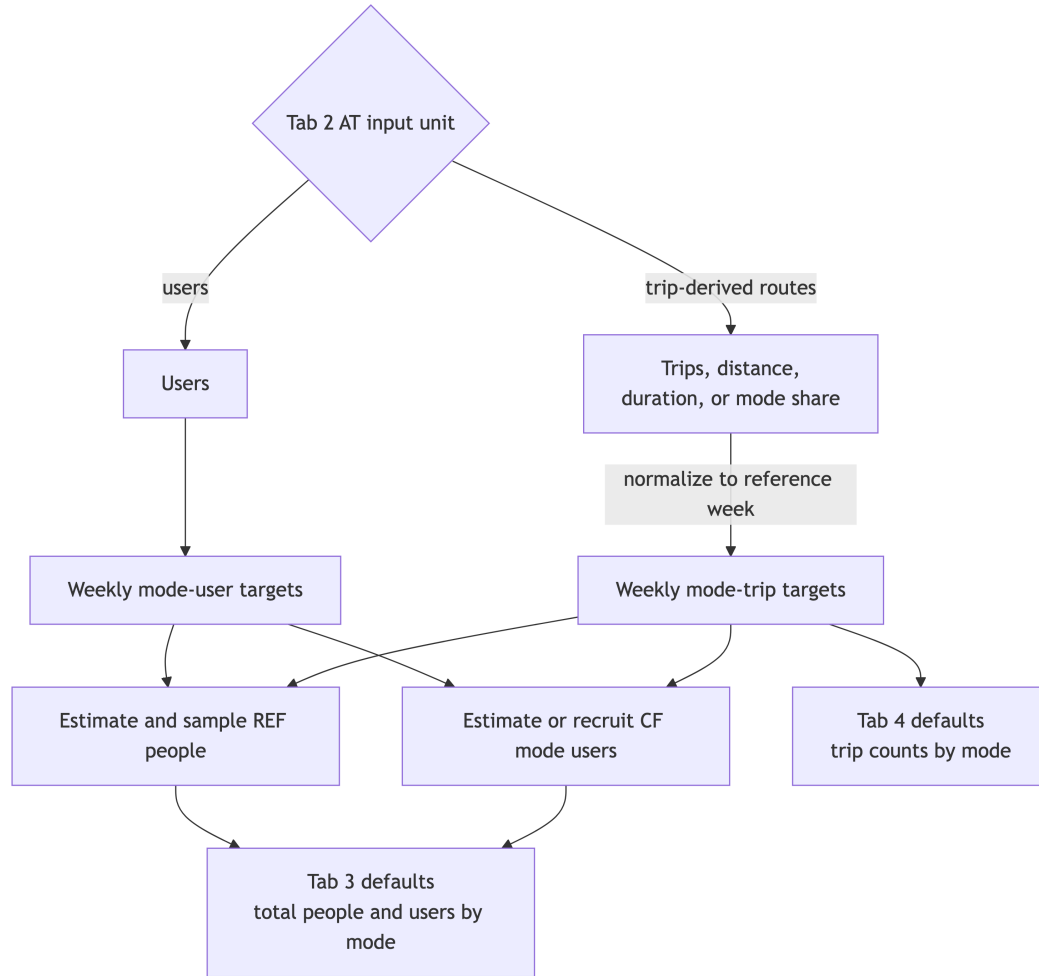


The basic and advanced interfaces use the same core conversions. The basic path accepts Tab 2 values and proceeds directly to calculation using generated population and trip assumptions. The advanced path exposes those generated values in Tabs 3 and 4 before calculation.



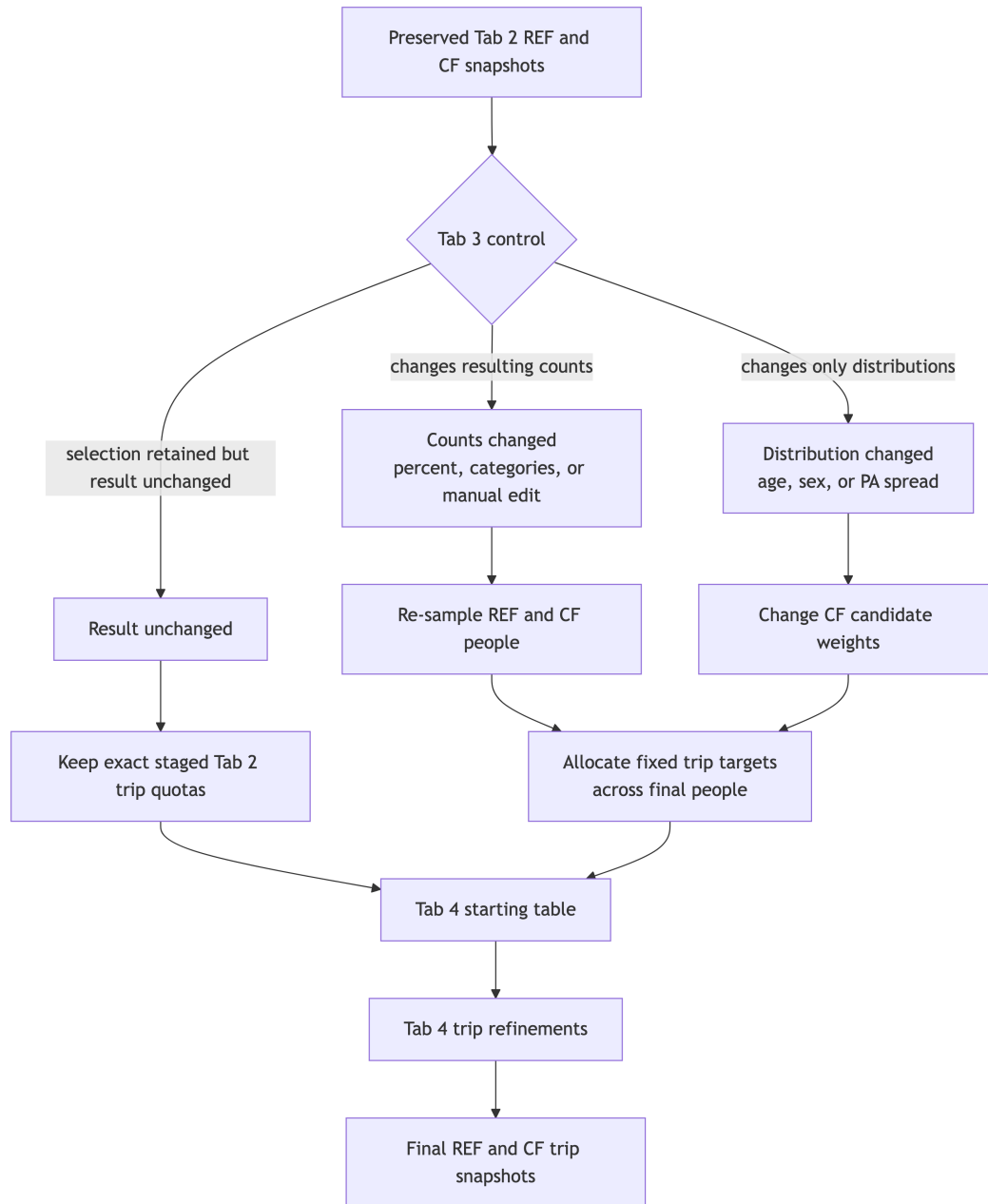
4.2.2 How Tab 2 units reach Tabs 3 and 4

All four Tab 2 input routes are normalized to weekly synthetic-row targets. This conversion creates the same downstream contract regardless of which unit the user understands best.



For users, Tab 3 receives the requested mode-user counts directly. For all trip-derived routes, Tab 3 receives the unique owners of the staged trips and an estimated common population boundary. If no Tab 3 population refinement changes that boundary, Tab 4 receives the exact staged Tab 2 trip targets.

4.2.3 What refinements change



For a trip-derived Tab 2 route, a genuine Tab 3 population refinement changes how the fixed trip quota is allocated, not the quota itself. If the final users own too few observed rows, HUB samples donor trip patterns with replacement and assigns them to those users. Merely selecting a refinement method that produces unchanged counts must not trigger re-scoping. HUB records actual triggers in `trip_refinement_defaults_report$tab3_rescope_trigger_fields`.

The complete geography is the donor population, but it is not necessarily the population represented by the appraisal. Filled Tab 2 reference fields first create a row-level **REF scoped** snapshot. Tab 3 reference totals, mode-specific user counts, and distribution anchors are recalculated from that snapshot rather than from the complete geography.

If Tab 2 specifies a mode-user count, HUB samples that number of observed users. If it specifies trip volume directly, or indirectly through distance, duration, or mode share, HUB converts the input to a weekly mode-trip target, samples active trips, and derives the corresponding mode-user count from the unique owners of those selected trips. When neither field changes, the geography-derived baseline remains the reference default. Explicit user and trip targets can both be supplied, in which case the trip scope is selected within the user scope.

The Tab 3 total is an estimate of the population represented by those Tab 2 volumes, not automatically the whole LAD. When no explicit total is supplied, HUB compares the requested selected-mode volume with the same volume in the source population:

$$N_{assessed} = N_{source} \frac{\sum_m V_{requested,m}^{REF}}{\sum_m V_{source,m}^{REF}},$$

where V is mode users for user-count input and weekly trip rows for direct or converted trip-volume input. Pooling the selected modes avoids making either walking or cycling the sole population anchor. People using multiple modes contribute to both sums consistently. HUB also reports each standalone mode estimate because strong disagreement between them indicates an internally atypical scenario.

When all applicable REF volumes are zero but at least one CF volume is positive, the REF equation cannot define a usable person boundary. HUB then applies the same equation to the CF target and uses that non-zero estimate for the shared staging scope. If a positive CF mode has no observable source rate, as may occur for e-bike, the full geographic person pool is retained instead. This fallback does not invent REF activity: REF mode-user and trip counts remain zero. It only keeps enough people and their observed trips available to construct the CF snapshot. A positive explicit population still takes precedence; a submitted zero population is overridden only when positive CF activity requires support. The rule applies equally to users, direct trips, and trip targets converted from distance, duration, or mode share.

Tab 2 counterfactual inputs are applied to a synthpop-only copy before Tab 3 is shown. This provides a no-health-model preview of the same CF population that will later receive Tab 3 and Tab 4 refinements. Values written into the profile remain defaults, not submitted inputs. A genuinely edited Tab 3 value takes precedence over Tab 2, while an unchanged rendered control that merely equals its generated default does not become a new constraint.

The left and right Tab 3 columns start as the staged REF and CF snapshots. Both remain editable. Basic percentage and age/PA-category controls scale both snapshots from preserved staged backups, retaining the REF/CF contrast while changing the overall assessment reach. HUB supplies separate absolute counts in `additional_data$ref` and `additional_data$cf`; the UI applies the selected categories to its own staged value. Manual values can still be entered independently after this operation. Advanced distribution sliders do not change table counts; they change the weights used to select rows for CF behavioral changes.

Operationally, MIAMA-UI first saves the Tab 2 controls into the profile and then calls `build_refinement_profile_defaults()` before rendering Tab 3. HUB returns the independently staged REF and CF defaults, backups, and category metadata; the UI renders and persists subsequent edits. After saving Tab 3 it calls `build_trip_refinement_profile_defaults()`, which applies the final Tab 3 population scopes and initializes Tab 4 trip values. It does not rebuild geography defaults, because doing so would replace the staged snapshots with the original baseline.

REF-to-CF default mirroring is likewise owned by HUB rather than repeated in modal-specific UI code.

The final Tab 3 person and mode-user counts take precedence at this handoff, while an authoritative trip-derived Tab 2 quota remains fixed. If the selected people own too few observed trip rows, HUB samples donor trip patterns with replacement and assigns them to the fixed users. The scope report records the added rows and allocation method, and HUB emits a warning. Any subsequent Tab 4 trip-count edit becomes the final trip target used for results without changing the accepted population counts.

The UI precedence for overlapping concepts is:

1. a filled, changed Tab 3 or Tab 4 value;
2. a filled Tab 2 value;
3. the geography-derived default.

4.3 Basic specification (Tab 2)

The basic interface offers alternative currencies for describing change:

Input route	Interpretation	Current modeling status
Number of mode users	Target number of weekly mode users	Implemented for all four modes.
Number of mode trips	Target weekly mode-trip count, with UI timeframe and denominator conversion	Implemented for all four modes.
Distance or duration	Target aggregate or per-person travel amount	Implemented for all four modes by conversion to a weekly trip-row target.
Mode share	Target shares by mode and a trip, distance, or duration denominator	Implemented for all four modes by deriving per-mode weekly trip-row targets.

Only one basic input route is intended to be active at a time. User and trip targets use the same counterfactual machinery as the advanced interface; the advanced controls add information about who changes and which trips change.

4.3.1 Assumptions that complete an appraisal

The assumptions-contract draft separates provided volume from the missing parameters needed to interpret it. Suggested frequency and trip-size values use assessed REF observations when available, otherwise the geographic source, then stored England rates and fixed fallbacks. Speed is a fixed assumption. At initial geographic loading there is usually no assessed REF yet, so source values apply. Defaults remain stable during subsequent sampling to avoid a circular calculation; their provenance and any user override are retained.

Distance and speed imply duration; duration and speed imply distance. These are never three independent assumptions. Saved overrides affect conversions, sampling preferences or new CF activity, not observed REF exposure. For changed trips, active minutes are sampled kilometres divided by the effective target-mode speed, multiplied by 60. PT refers to the walking access leg. Newly assigned users retain donor variation, with its expected activity scaled by the effective weekly frequency and duration; the existing accounting avoids adding the same user-sourced trip activity again.

The cards omit inputs already provided elsewhere and unchanged distributions. An explicit advanced distance refinement supersedes a simple distance mean. Accepted person/trip counts remain authoritative: editing an assumption does not by itself erase them. A preferred mean is a sampling weight, not a guarantee that the realized sample has exactly that mean. Completed results retain the effective assumption values, defaults and origins for review.

The implementation and remaining provenance/presentation questions are described in [Assumptions that complete an appraisal](#). The stored England values retain earlier UI rates; they are not freshly estimated national statistics. This draft requires the matching HUB and UI assumptions branches.

4.3.2 Converting distance, duration, and mode share

The behavioral model uses one canonical unit for trip-volume manipulation: the number of synthetic trip rows in a reference week. Alternative Tab 2 inputs are therefore converted before either REF scoping or CF sampling.

For mode m , a distance or duration target is first converted to kilometres or minutes and then to one week. A per-person input is multiplied by the applicable scenario population N_s ; a total input is already an aggregate. Let $A_{m,s}$ be that weekly aggregate and let $\bar{a}_{m,REF}$ be the positive mean distance or duration among reference trips of mode m (or the effective completion assumption, when using the assumptions contract). The trip-row target is

$$T_{m,s} = \text{round} \left(\frac{A_{m,s}}{\bar{a}_{m,REF}} \right).$$

For mode share, let $p_{m,s}$ be the requested percentage and D_s the selected trip, distance, or duration denominator. The mode-specific amount is

$$A_{m,s} = D_s \frac{p_{m,s}}{100}.$$

If D_s is a trip total, $A_{m,s}$ is rounded directly to a trip-row target. If it is total distance or duration, the first equation is used. This makes mode share a per-mode sampling quota rather than a separate row-selection algorithm.

The converted targets pass through the same constrained trip sampler as direct trip-count inputs. Target volume controls how many active-mode rows are needed; Tab 4 distance distributions and diversion rates control the relative probability that eligible rows are selected. The shifted-versus-induced rule controls whether those rows come from existing non-active trips or copied active-trip donors. Resulting changes in active minutes are converted to MMET exposure through the standard health pathway.

In a zero-REF scenario, owners of the trip rows actually shifted or induced are recorded as CF mode users even when integer rounding of the preferred new-user percentage initially selects no recipients. This keeps the Tab 3 population summary consistent with the realized CF trip snapshot without adding individual activity exposure a second time.

The values last displayed or entered in the Tab 4 table are the final trip-volume targets. Generated starting values already carry forward or derive from Tab 2, so accepting them preserves that scenario; an explicit Tab 4 edit supersedes it. Diversion settings are independent of this precedence rule because they influence source-trip selection rather than trip volume.

For distance, duration, and mode-share input routes, the Tab 2 aggregate is the provenance used to derive Tab 4's starting trip counts. An explicit Tab 4 count supersedes that derived target. HUB does not then force the original aggregate kilometres, minutes, or shares through a reverse adjustment of people or trips; it reports the values realized by the final sampled trips. Consequently, a Tab 4 change may alter trips per user and the final measured mode share while the accepted Tab 3 population and mode-user counts remain fixed.

Tab 4 defaults are measured from the separate staged REF and CF snapshots created after Tab 2. With no Tab 3 population change, these snapshots pass through unchanged. When Tab 3 changes the population scope, HUB re-scopes each snapshot independently and allocates its carried trip target across the final users. This preserves the scenario-specific behavior already introduced by users, trips, distance/duration, or mode-share inputs.

Tab 4 presents one source-mode diversion pie for each assessed target mode. A cycling pie, for example, describes the previous modes of existing trips that will shift to cycling; cycling itself is excluded. The pies apply only to shifted trips, while `induced_trips_percent` controls newly created trips separately. HUB uses these target-specific shares as donor sampling weights and records both requested and realized source-mode distributions. Defaults come from configured assumptions where available, otherwise from eligible local donor-trip composition. The pies are count-based conditional distributions and do not need a separate trip, distance, or duration denominator.

This same fallback covers appraisals that stop at the basic Tab 2 mode-share input. In the absence of a submitted Tab 4 pie, HUB uses a configured target-specific split where one exists; otherwise it uses eligible utilitarian reference donors outside the assessed active modes. This avoids order-dependent competition between basic active-mode targets. It is a neutral sampling fallback, not an empirical estimate of intervention diversion behavior.

The conversion is an expectation-based approximation. It preserves the requested aggregate when sampled rows resemble the reference mean, but it does not enforce an exact kilometre or minute total after finite, constrained sampling. HUB records all conversion inputs and derived row targets in `reference_scope_report$tab2_input_conversion` and `counterfactual_report$tab2_input_conversion` so requested and realized values can be compared.

An average per trip is not a complete volume specification because it supplies no number of trips. HUB therefore rejects `average_per_trip` in this pathway; it belongs in Tab 4 as a characteristic constraint unless paired with a second volume input. Mode-share distance and duration denominators currently use canonical reference-week units of km/week and minutes/week. Explicit UI unit and timeframe controls remain a schema refinement if other interpretations are to be supported.

4.4 Refined population specification (Tab 3)

The population equivalents inferred from Tab 2 are shared by the optional Tab 2 basic population modal and the Tab 3 handoff. HUB writes the staged counts to both the `*_basic` and `*_advanced` profile fields. Consequently, opening the basic modal after supplying mode shares should show the pooled Tab 2 population estimate and the corresponding REF/CF mode-user counts, not an unrelated copy of the full geographic source defaults. Non-user input routes retain the same total population in REF and CF while allowing mode-user counts to differ.

For each assessed mode, Tab 3 can provide:

- a target number of affected mode users;
- a target mean age and proportion male;
- a target mean weekly PA exposure and proportion male within the PA display; and
- a UI option describing current versus new users.

Direct user-count inputs are authoritative and therefore supersede the current/new-user percentage. The percentage is applicable when user counts must instead be inferred from trip, distance, duration, or mode-share inputs. HUB divides the absolute change in weekly trips by the profile's mode-specific weekly trips-per-user assumption to estimate a trip-equivalent affected count. Existing users of that mode absorb the current-user share of additional trips; only the selected percentage is added to the mode-user population. E-bike uses current active-travel users as its proxy current-user pool because no reference e-bike users are observed. This changes the mode-user scope shown in Tab 3, while health exposure remains attached to changed trip minutes to prevent double counting.

When an explicit counterfactual mode-user count exceeds its reference count, all current mode users are retained and the difference is recruited from eligible baseline non-users across the full retained geographic source. The realized current/new split is therefore determined by the explicit counts and may exceed the preferred new-user percentage. HUB reports that realized split and adds a non-fatal note when this occurs. Too few distinct donors trigger eligible person replacement; an absence of eligible donor evidence still fails.

The aggregate CF population total is a presentation summary. Row-level behavioral changes are controlled by the mode-specific CF user targets. People who remain behaviorally unchanged are not removed from health denominators merely to force the displayed aggregate CF total.

When Tab 3 is initialized, its current/new-user slider uses the realized split from the CF snapshot staged from Tab 2. For multiple assessed modes, HUB pools the mode-user memberships represented by each change. This is a mode-membership summary rather than a count of mutually exclusive people because one person may use more than one mode.

Reference distributions are represented as a 5 x 2 table: five configured age or PA categories crossed with sex. The app displays the table as paired bars. Moving a mean or proportion slider does not transmit a new table to HUB. Instead, both UI and HUB reproduce it from the same reference bars and scalar counterfactual slider values.

For the numeric dimension, exponential tilting shifts the reference category percentages toward the requested mean while retaining the reference nonzero support. For the binary dimension, the requested proportion replaces the reference marginal. The two counterfactual marginals are combined using an outer product, which assumes independence between the numeric category and the binary grouping. If neither slider changes, the reference table is returned exactly.

Current configured categories are:

- age: 18–29, 30–39, 40–49, 50–59, and 60+;
- PA: sedentary, low, moderate, high, and very high, using provisional MMET cutoffs in configuration.

4.5 Refined trip specification (Tab 4)

For each assessed mode, Tab 4 can provide:

- a target number of weekly active trips;
- a target mean trip distance;
- the percentage of additional trips that are newly induced rather than shifted from existing travel; and
- the percentage of new active-mode shifts expected to come from car.

Trip spread bars use five configured distance categories crossed with utilitarian/recreational purpose for descriptive display. The same slider redistribution method is used for reconstructing internal counterfactual distance targets. Purpose is not a separate sampling target in the intended contract: shifted existing trips must be utilitarian, while newly induced trips are normally recreational. The legacy purpose input route should therefore be retired in coordination with MIAMA-UI rather than developed into a second mechanism control. The configured distance groups are 0–2, 2–5, 5–10, 10–30, and 30+ km.

When Tab 4 is initialized, its induced-trip slider uses the realized proportion of added CF trip rows that were induced rather than shifted. This is deliberately separate from the Tab 3 people split. If no trip addition has yet produced a realized mechanism split, the effective Tab 3 current/new-user percentage is used only as the initial Tab 4 assumption.

5 Counterfactual sampling and redistribution

5.1 General design

The complete filtered LAD synthetic population is retained as the baseline source pool. User-entered reference counts define an **appraisal snapshot** within that pool. HUB records separate scope flags for assessed people, mode-specific users, and mode-specific active trips. Selecting the reference snapshot does not alter behavior, MMET exposure, or health trajectories; it only sets the system boundary for the appraisal.

Counterfactual data begin as a one-to-one copy of this flagged baseline. CF scope initially equals REF scope. New users are sampled first from baseline non-users inside that assessed population. If more are needed, baseline non-users from the retained LAD source pool are recruited and added to CF scope. When capacity is insufficient, source people and their trip patterns are copied with unique appraisal IDs. Retained donor IDs link each copy to its source health history; REF and CF use the same appraisal ID. Existing trips can be switched only for people in CF scope, while induced trips are assigned to CF scope. All behavioral changes are made only in the CF copy; REF scope and behavior remain unchanged.

The random seed is fixed by the calling workflow, making a run reproducible for the same data and inputs. Existing trip records cannot be shifted repeatedly. Eligible person donors and trip patterns may be reused when distinct source capacity is insufficient. Copies have unique appraisal identities and retain their source-donor links; each copied person’s health history is loaded once. With unit person weight, each copy represents one additional appraisal person, but it does not add independent source evidence.

The distinction between an appraisal boundary and the source pool is important. REF uses observed donor patterns: its mode-user counts must fit inside REF total population, but eligible donors can be reused for larger requests. CF is not subject to the smaller REF population ceiling. It can recruit eligible non-users from the rest of the retained geography and reuse donors when needed. No eligible evidence or incompatible margins still cause an error. HUB reports such input incompatibilities separately from programming errors, with requested and available counts and the implicated profile fields.

Age and PA category selections are treated as hard inclusion criteria. If a previous REF table value is stale, HUB reconstructs scenario-specific total and mode-user values from the absolute category counts stored in the profile during Tab 2 -> Tab 3 staging. This duplicates the UI calculation deliberately and removes dependence on browser update timing. If stored counts are unavailable or exceed distinct eligible capacity, HUB reuses eligible donors to preserve accepted targets. It does not relax selected categories. Donor reuse and proxy use are recorded in the replication report. A cycling proxy can supply labelled reference e-bike donors when native e-bike observations are absent.

The final category-refined snapshot is currently reconstructed from eligible source rows using the staged category totals and a reproducible seed. It is not guaranteed to preserve literal row identity from the intermediate Tab 2 draw. This distinction should be revisited if variance analysis indicates that exact nested sampling materially changes results.

5.2 Changing the number of users

If the counterfactual target exceeds the reference user count, HUB samples baseline non-users and marks them as **new_users**. It exhausts eligible candidates already inside the assessed population before recruiting from the rest of the retained LAD. If the target is lower, HUB samples existing users and marks them as **ex_users**; those people remain members of CF scope.

A CF target is not capped by the smaller scaled REF population. Its practical capacity can grow through reuse of baseline non-users that remain eligible under the selected population refinements. If no eligible donor evidence exists, HUB returns a structured appraisal-input error with requested and available counts.

Candidate probabilities combine the available mode-specific Tab 3 constraints:

$$w_i = w_{sex,i} \times w_{age,i} \times w_{PA,i}.$$

The category percentages reconstructed from the sliders are used directly as relative sampling weights. They are not exact quotas and are not currently calibrated for the prevalence of each category in the candidate pool. If too few candidates have positive combined weight, all positive-weight candidates are selected first and the unavoidable remainder is sampled uniformly. The counterfactual report records this relaxation.

New users receive a weekly mode-activity value sampled with replacement from observed current users of that mode; the configured fallback is one hour per week if no donor value exists. Ex-users' mode activity is set to zero. This changes the person's MMET exposure.

When trips are available, new-user status also triggers mode shifts among that person's plausible non-active trips. The number is based on the configured weekly trips-per-user assumption; observed trip rates among current users are a fallback. For ex-users, their active trips are switched away from the active mode, currently to car.

Warning

The age and PA spread targets constrain **which candidates change status**. They do not impose exact distributions on all final counterfactual users. In addition, the PA constraint is based on candidate PA before newly sampled mode activity is added, so the final PA distribution of new users can differ from the displayed target.

5.3 Changing the number of trips

For an increase, the default mechanism treats 90% of additional active trips as mode shifts and 10% as induced recreational trips. The explicit parameter is `induced_trips_percent` in the submitted profile, optionally overridden by a mode-specific field such as `induced_trips_percent_walk`. If neither is provided, HUB uses `cfg$counterfactual$trips$induced_trips_percent_default`. This mechanism parameter is distinct from both trip purpose and the new-user percentage, even when defaults happen to use the same value.

Mode-shift candidates are existing utilitarian trips. By default, trips already active in another assessed mode are excluded. A configured source-mode distribution can explicitly make them eligible, as for e-bike shifts from cycling or PT. Candidate distances are limited to 120% of the maximum observed reference active-mode distance to avoid implausible switches. Their sampling probabilities reflect the requested distance distribution or, when only a mean is available, a smooth preference for distances near that mean. The requested car-source share rebalances source-mode candidate pools while retaining the distance weights. For e-biking, the current configuration assigns the non-induced shifts equally across cycling, PT, and car because no England-derived split is yet available. This source-mode assumption is distinct from the cycling donor profile used to impute e-bike characteristics. A UI car-share value overrides the configured car share and leaves cycling/PT in their configured relative proportions. The realized share can differ in finite or constrained samples and is reported.

Selected trips retain their observed raw distance and duration; the target active-mode fields are populated from those raw values. For induced travel, HUB copies observed active-trip donor rows, assigns them to people in CF scope, creates new trip identifiers, marks them recreational, and retains the donor trip weight.

For a decrease, sampled active trips are switched to the configured alternative mode, currently car. Rows are not deleted because the underlying travel need is assumed to remain.

i Note

Legacy scalar and spread-purpose controls may still be present in the profile schema. They are not authoritative sampling inputs. The explicit `induced_trips_percent` controls the mechanism split, from which purpose is inferred.

5.4 Count units and represented totals

The current appraisal contract is deliberately direct: one synthetic person row is one user record and one synthetic trip row is one trip record. Reference defaults, UI trip targets, counterfactual sampling, result trip totals, and mode shares therefore do not apply `weight_tripXhh`. Distance and duration totals sum the corresponding values across included trip rows. Source weights remain in the data for future methodological work, but they do not change current count targets. Health-result population expansion is a separate configured operation.

6 What each refinement currently controls

UI information	Stored/reconstructed quantity	Current effect on counterfactual data
Mode-specific user count	Weekly target users	Adds new-user or ex-user statuses among existing people.
Mean age + sex proportion	Five age categories and sex marginal	Weights candidate people changing status.
Mean PA + sex proportion	Five PA categories and sex marginal	Weights candidate people using their pre-change PA.
Current/new-user percentage	Scalar profile value	For trip-derived inputs, divides additional mode users between existing active-travel users and people with no current active travel; explicit user counts take precedence.
Mode-specific trip count	Weekly target trip rows	Shifts existing trips and adds induced recreational trips.
Mean trip distance	Five distance-category target or smooth mean preference	Weights plausible trip candidates.
Induced-trip percentage	Share of added trips that do not replace an existing trip	Splits additions between induced recreational rows and shifted existing utilitarian rows.
Purpose display	Descriptive utilitarian/recreational distribution	Derived from the trip mechanism; legacy purpose inputs are scheduled for coordinated retirement.

UI information	Stored/reconstructed quantity	Current effect on counterfactual data
Source-mode diversion	Target-specific expected source shares among shifted trips	Uses one source-mode pie per assessed target mode; unavailable donor pools are omitted and remaining shares renormalized.
Basic distance/duration target	Weekly aggregate km or minutes, total or per person	Converted to a mode-specific weekly trip-row target using the reference mean per active trip, then passed to constrained sampling.
Basic mode share target	Mode percentage plus a trip/distance/duration denominator	Converted to a per-mode weekly trip-row quota, then passed to constrained sampling.

7 Validation, uncertainty, and remaining work

The current implementation makes assumptions visible in `counterfactual_report`, including requested and realized counts, active sampling constraints, fallback sampling, induced trips, diversion shares, and MMET components. This supports mechanical checking, but it is not a substitute for empirical validation.

7.1 Priority methodological TODOs

1. **Current/new-user validation:** compare the trips-per-user conversion and realized current/new allocation with empirical intervention evidence, especially where one candidate pool is too small and the other fills the shortfall.
2. **Candidate-weight calibration:** decide whether category weights should be calibrated as target share divided by candidate prevalence, or replaced by exact/near-exact quota sampling. Test on large and sparse geographies.
3. **Final-distribution validation:** report requested versus realized age, sex, PA, distance, purpose, and source-mode distributions after all changes, not only the constraints supplied to the sampler.
4. **Purpose simplification:** retire legacy purpose controls in UI and HUB, retain the explicit induced-trip percentage, and report the realized utilitarian/recreational split derived from mechanism.
5. **PA targeting semantics:** decide whether PA categories refer to pre-change activity, post-change total activity, or mode-specific added activity, then align sampling and UI wording.
6. **Trip weights:** validate the current one-row/one-record interpretation and decide whether a future reporting layer should expose survey-weighted or population-expanded estimates separately from sampling targets.
7. **Alternative-volume calibration:** compare requested versus realized aggregate kilometres/minutes after constrained row sampling; decide whether exact or near-exact aggregate

calibration is required. Add explicit UI unit/timeframe controls for mode-share distance and duration denominators if canonical weekly km/min assumptions are insufficient.

8. **Short-term effects:** implement a time-limited intervention effect rather than applying the MMET change to every health cycle.
9. **Category and intensity review:** validate age, distance, PA cutoffs and MMET intensity factors against agreed evidence and document provenance.
10. **Mode proxy validation:** validate the provisional cycling-to-e-bike factors (distance 1.30, speed 1.20, duration 1.30 / 1.20; frequency unchanged) and the configured PT access-walking assumptions when mode-specific evidence becomes available.
11. **Uncertainty analysis:** quantify Monte Carlo variation from sampling and parameter uncertainty; a fixed seed currently provides reproducibility but not uncertainty intervals.
12. **External validation:** compare reference summaries, modeled behavioral changes, and health-response magnitudes with independent observed data and established appraisal tools.

7.2 Recommended checks for each appraisal

- Confirm that reference user/trip counts and spread plots are credible for the selected geography and are not using a sparse-data fallback.
- Compare requested and realized counterfactual changes in the report.
- Inspect the number of relaxed sampling selections and induced trip rows.
- Check the individual mode-specific MMET components before reviewing health outcomes.
- Confirm sign conventions: internal delta is counterfactual minus reference; prevented outcomes are reference minus counterfactual.
- Treat very small samples and unsupported modes as development tests, not substantive estimates.

8 Appraisal population summary

Appraisal inputs describe real people: one selected individual record represents one person. Health totals and population denominators use unit person weight by default. Geographic expansion factors are retained as source metadata only; they neither reduce sampling targets nor multiply appraisal outcomes. Thus a request for 1,000 users still samples 1,000 user records. Default counts describe the record-sized example appraisal, not the entire source geography. Existing saved configurations and results must be rebuilt to adopt this convention.

The Tab 5 population summary describes the final counterfactual appraisal: total people including non-users, and users of walking, cycling, e-biking and public-transport active access. It counts assessed individual records without applying geographic expansion weights to these summary counts. It does not count the entire donor population or restrict the population to rows with health estimates. Mode groups overlap, so their sum is not the total population. These are realized counts rather than a second population estimate or sampling input.

The results report combines the appraisal inputs, health impacts and assumptions in a single HTML or Word document. It records the calculation inputs at the time results are built, including the accepted population and trip values, rather than reading subsequently edited controls. Realized travel totals are shown separately from requested values. Completion assumptions include their

units and source (assessed reference, source population, stored national rate or fixed fallback) where recorded by the assumptions catalogue.

Submitted settings, effective calculation inputs and configured defaults are distinct sections: saved inactive fields are not evidence of use, and defaults can be overridden. In particular, a requested new-user or induced-trip percentage is not a measured realization of that split. The report is not yet a complete donor-selection audit and does not quantify sampling or structural uncertainty. It contains no individual-level records. The recorded export selection applies to the health tables, while headline metrics describe the full appraisal; later on-screen filters do not modify this export snapshot. # Implementation traceability

This description is based on the current implementation, principally:

- R/reference_data_extract_reference_ui_values.R
- R/tab2_input_conversion.R
- R/reference_data_spread_functions.R
- R/counterfactual_data_sampling_functions.R
- R/counterfactual_data_apply_ui_values.R
- R/counterfactual_data_apply_health_outcomes.R
- R/results_data_prepare.R
- R/config.R and R/constants.R

The companion slide deck is docs/methodology_slides.qmd. The package README remains the developer-facing integration reference; where terminology differs, the implementation and the status statements in this methods document take precedence until the discrepancy is resolved.